

Recommended Assessment

Camera

Exploring Camera Sensors

1. What do you notice about the motion of the object?
2. Write down the resolution, camera intrinsic and extrinsic parameters. Do you think these values will change with a different resolution? Note the serial number of your device too.

Resolution	Intrinsic Parameters	Distortion Coefficients

3. What is the focal point of the camera? How do calibrated focal length and manufacturer provided focal length vary
4. Change the environment (close a window, dim lights or add another light source). Were you still able to detect the color in the camera image?

Real Life Scenarios

5. You are tasked to select a camera for incoming part inspection. Parts will be coming in from a conveyor belt and depending on the machine vision pipeline the parts will be rejected or not. What type of shutter and lens combination should the camera have?
6. A safety monitoring system uses a camera system to perform motion detection and detect intruders. You're asked to make sure the system works well in light and dark conditions. What are some considerations when deciding the camera system?

Camera Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	